

Nike, Inc. Case Study:

I do not agree with Joanna Cohen's weighted average cost of capital calculation of 8.4% for multiple reasons. First, Joanna used the book values instead of the market values of Nike's debt and equity, which is what investors are actually looking at. This mistake shows itself in Exhibit 5 of the case study when Cohen claims that debt makes up 27.0% of total capital and equity accounts for the remaining 73.0%. By using market value of debt and equity rather than book value, Cohen would instead find that debt only makes up 10.0% of total capital while equity makes up 90.0%, which is a significant change. She also calculated the cost of debt by dividing interest expense by the average debt balance, which is the incorrect method in finding this number. Instead, Joanna should have used yield-to-maturity on the company's public debt. Last, she used the 20-year treasury yield, which is 5.74%, as the risk free rate in her solution. What she should have used was the 10-year yield, 5.39%. The 10-year treasury yield is the common benchmark used in corporate investments and will result in a more accurate estimate on Joanna's part.

To correct Joanna's WACC estimation, you need to use the yield-to-maturity method on Nike's traded bond using the 6.75% coupon, 20 years to maturity, and price of \$95.60. The result is a pre-tax cost of debt of 7.17% before applying the 38% tax rate, and 4.44% afterwards. Using CAPM to find the cost of equity, you take the 10-year Treasury yield of 5.39% for the risk-free rate, Nike's average beta of 0.80, and the risk premium of 5.90%. This results in a cost of equity of 10.11%. Then, to solve for the correct WACC, multiply the after-tax cost of debt by the market value of debt and add it by our cost of equity multiplied by the market value of equity to find a WACC of around 9.56%. This result is an increase from Joanna's finding of 8.4%.

We have already calculated the cost of equity using the CAPM method, finding an estimate of 10.11%. Another method of finding the solution to this is using the Dividend Discount Model. Using this method, you divide the expected dividend of the next year by the current stock price, then add the constant growth rate. Exhibit 4 shows us that Nike paid \$0.12 per quarter throughout 2000, giving us \$0.48/share annually as the current dividend. Next year's expected dividend is found to be around \$0.51, calculated by multiplying \$0.48 by the dividend growth rate of 5.5% (or 1.055). After plugging these values into the DDM formula alongside the current stock price of \$42.09 to find the cost of equity to be 6.70%. This means that the cost of equity found by CAPM is over three percent greater than that found by DDM. The advantages of CAPM when finding cost of equity include that it accounts for market risk by involving beta in its formula and uses observable data to achieve its solution. It is also the most commonly used method. Some disadvantages, however, are that beta may not entirely reflect future risk and the method assumes potentially unrealistic circumstances such as a linear relationship between market and stock returns and that investors can borrow at a risk-free rate. An advantage of using the Dividend Discount Model is that it assumes less than CAPM and is more simple in the sense that it directly ties the cost of equity to what shareholders actually receive in dividends. On the other hand, DDM is sensitive to changes in growth rate and the growth rate assumption

that comes with it is unreliable due to Nike's very low dividend yield of around 1.14% as calculated by dividing the annual dividend by the stock price ($\$0.48/\42.09).

Kimi Ford, regarding an investment in Nike, should recommend buying stock in the company. With a corrected 9.56% weighted average cost of capital (which is less than the 11.17% discount rate when the company is valued at \$42.09), the stock is shown to be undervalued. Exhibit 2 presents a \$55.68 equity value at a 9.5% discount rate, which shows upside from the current price. In addition, Nike's revenue returned to growth in 2001 at 5.5% and margins have been steady at around 40% as shown in Exhibit 1. A strategy was also outlined by the company on June 28 with the goal of 8-10% revenue growth and at least 15% earnings growth, stated on page one of the case study. Despite potential risks and mixed reactions from analysts, Nike's stock was shown to have increased above the S&P after this meeting occurred, seen in Exhibit 4. For NorthPoint Group, a mutual fund focused on large-cap stocks, Nike at a \$42.09 share price is trading below its value and is a high potential investment opportunity.

California Pizza Kitchen Case Study:

As the CFO of California Pizza Kitchen, Susan Collyns can recapitalize the company's capital structure to facilitate success. The company's stock currently does not reflect the performance of the company (being undervalued compared to analyst targets in the \$37-40 range) having dropped 10% to \$22.10 in June 2007 despite second quarter revenues up over 16% and sales growth over 5%. Collyns can look to take advantage of CPK's tax benefits of debt and improve return on equity. Apart from the capital structure, she can maintain the cost controls like labor and food costs that have seen the most success. Labor costs have stayed consistent at 36.3% of revenues and food costs have remained at 24.5% across multiple years. She also needs to make sure that the \$85 million in planned expenditures for 2007 is funded in a balanced manner that results in both business growth as well as shareholder return value. Being open with investors is an unquantifiable aspect of this project that is extremely important as well in order to keep them aware of the company's new approach and how it affects its growth and future value.

In Exhibit 9 of the case study, leverage plays a significant role in both return on equity and cost of capital. ROE increases at every level of debt due to equity decreasing faster than earnings do. At a debt of zero, ROE is 9.0%. When debt goes up to 10%, ROE in turn rises to 9.5%. 10.2% ROE at 20% debt, and 11.1% at 30% debt. This occurs because the use of debt causes increases in returns going to equity holders due to leverage. For cost of capital, CPK's after tax cost of debt is 4.16%, found by multiplying 6.16% by 67.5% ($1 - 0.325$) tax rate. This makes it significantly cheaper than financing equity. The company's weighted cost of capital decreases as CPK swaps debt for equity, in turn increasing firm value. In addition, adding debt increases the company's beta and makes equity more volatile, adding another reason to keep leverage moderately rather than pushing to the higher end.

The tax deductibility of interest is a strong argument toward having California Pizza Kitchen take on debt. Interest payments lower taxable income while dividends do not. Because of this, every dollar of interest paid by CPK saves \$0.325 in taxes. In Exhibit 9, total income taxes decrease from \$9,755 at zero debt to \$8,400 at 30% debt, an enormous saving benefit that the company has been missing out on for years due to not taking on any debt and fully using equity instead, which is unnecessary anyway since the company has stable and predictable cash flows from over 180 restaurant locations. This presents a government subsidy on debt financing that has been ignored by California Pizza Kitchen for years.

I would recommend California Pizza Kitchen takes on a 20% debt-to-total capital structure. If this is done, return on equity increases to 10.2% from the original 9.0%. In addition, the annual tax shield is just over \$900,000 and the debt needed is around \$45 million, which is doable having \$75 million in credit line. CPK already carries \$154.3 million in lease obligations for the next five years, which is another form of leverage. Going straight to 30% debt is too aggressive because the company's financial obligations are already higher than the balance sheet shows. While the 10% and 30% are not aggressive or too aggressive, respectively, to immediately jump into, the 20% structure allows the company to buy back shares at a price cheaper than analysts think the stock should be while maintaining the flexibility to fund new restaurant locations. This approach will allow Rosenfield to be safe with the company's staying power while adjusting CPK's use of debt to actually make the company more valuable. California Pizza Kitchen is in a position to take on moderate debt and see little risk while also checking off multiple boxes when it comes to company goals, making it an ideal choice to make.